

Underwater Mass Spectrometry Advancements for Real-time Geochemical Analysis of Deep-sea Cold Seeps



September 18, 2025

Dr. Ryan Bell

Beaver Creek Analytical, LLC (BCA)

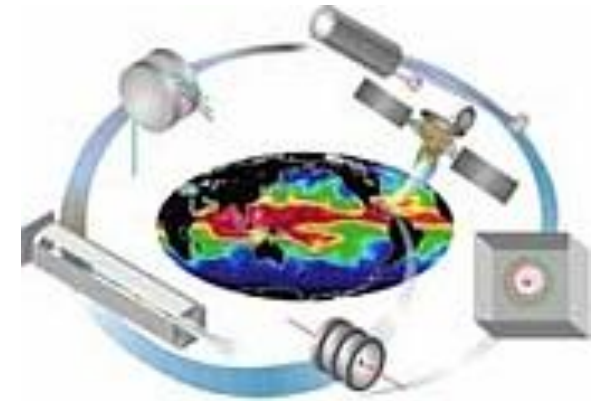
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Dr. Torben Gentz - AWI-Bremerhaven

Mr. Malte Höhn - AWI-Bremerhaven

Workshop on Harsh-Environment Mass
Spectrometry



With content from our partners at:



Outline

- Company Introduction
- In Situ Sensing
- Seeps Description
- UMS Description and Upgrades
- Cruise Results

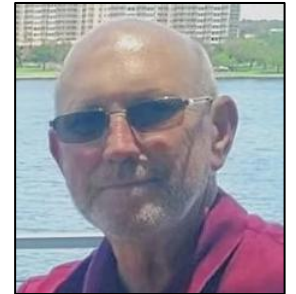


Company Introduction



Who we are...

- **Managing Owner**, Dr. Ryan Bell
 - Doctorate in Chemical Oceanography (USF)
 - 20+ years of portable instrument development (chemical, electrical, mechanical, firmware, software, deployment, data analysis, reporting)
- **Mass Spectrometry**
 - Dr Tim Short
- **Instrumentation and Geochemistry**
 - Dr Torben Gentz, Malte Höhn



What we do...

- Products for real-time, in situ chemical analysis
 - Dissolved gas sensors, **Underwater mass spectrometry**, others
 - Subsea accessories (pumps and valves)
 - Sales/rental
- Services - Offshore/onshore Campaigns
 - **Consulting**
 - Offshore seep characterization **campaign management**
 - Shipboard **Analytical Chemistry Services**
 - Hydrocarbon composition and isotopes
 - Sensor-triggered discrete sample collection
 - **Software**
 - **Automation** and data presentation of subsea analysis system
 - Mobile lab interface for **real-time emissions** data
 - **MS interface** for portable chemical analyzers (MKS, Detect-ION)
 - Contract R&D
 - Custom sensors
 - Targeted sample collection (porewater, bubbles, seep-fluid)

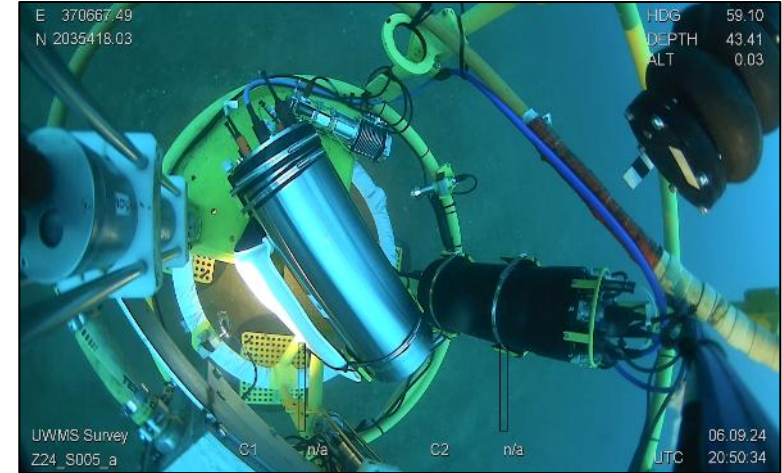


In Situ Sensing



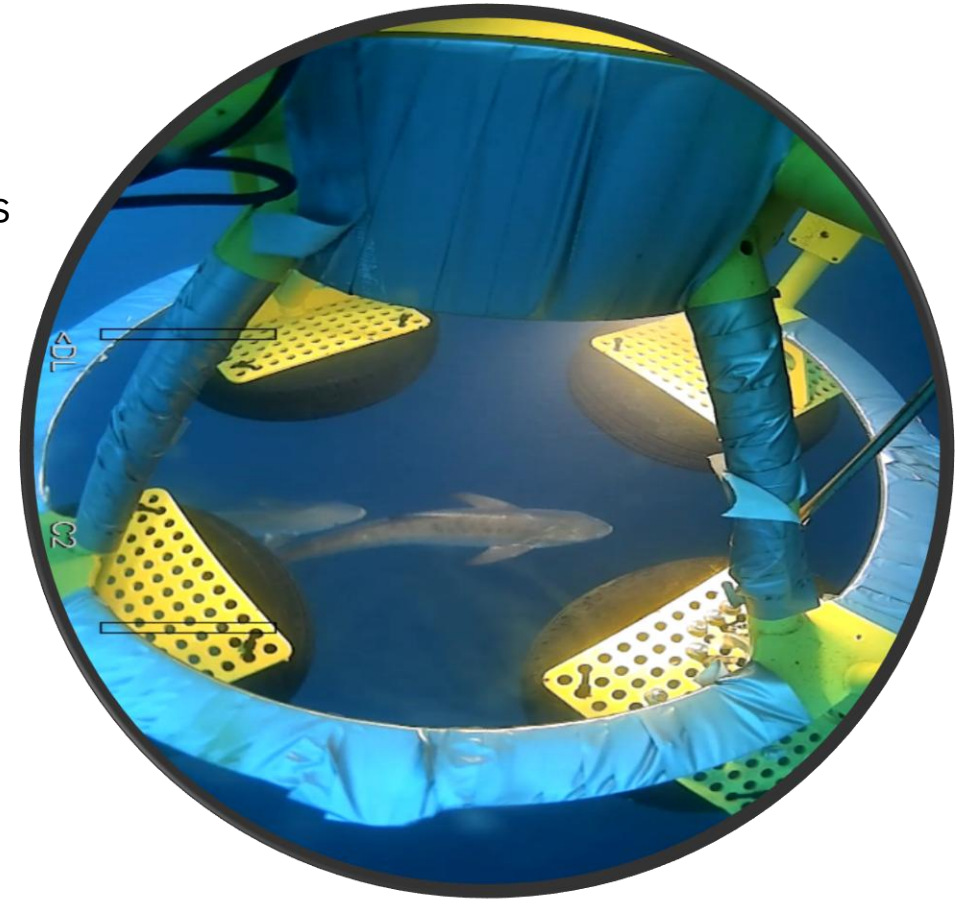
Benefits of In Situ Data

- Reduced sample **contamination**
 - Critical for volatile and reactive analytes
- Increased sampling **speed/density**
 - Broad, dense coverage aids data-interpretation
- **Gradient** mapping
 - Capturing many points along a gradient ensures extreme transients or local minima/maxima are not overlooked



Benefits of Real-time Data

- Intelligent, **adaptive sampling** to optimize ship-time
 - Low-interest areas: reduce sampling, save ship-time
 - High-interest area: increase sampling, improve chemometrics
- **Rapid response** to emergent threats
 - Rapidly deployment of response equipment
 - Reduce false positive responses
- **Self-directed** platforms
 - AUVs can execute targeted sampling paths independently
- Sensor-triggered **discrete sampling**
 - Sensor data can trigger collection of discrete samples for shipboard or accredited lab analysis

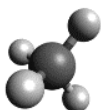


Underwater Mass Spectrometry - Overview

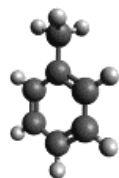


Simultaneous Detection of Multiple Analytes

Group	Example Sources	Examples
Atmospheric Gases	Air-sea interface	Nitrogen, oxygen, argon, carbon dioxide, noble gases
Aliphatic hydrocarbons	Natural gas, crude oil, refined products (e.g. hydraulic fluids)	C1-C7 alkanes (Methane, ethane, propane...) C5-C7 cycloalkanes (Cyclopentane...)
Aromatics	Fossil fuels, industrial	Toluene, benzene, xylene, ethylbenzene
Sulfides	Thermal vents, anoxic bodies (sink holes, fjords, basins, brine-pools), blooms	Hydrogen sulfide, methanethiol, dimethyl sulfide
Halocarbons	Treated water, hypersaline bodies,	Chlorobenzenes, chloroform, bromoform, many more
Non-volatiles (via reagent addition)	various	DIC, total hydrogen sulfide, various break-down products by oxidation, or chlorination
Larger MW Compound (via thermal release)	Persistent organic pollutants, crude components, drugs, toxins	PCBs, pesticides, dioxins, furans, flame retardants, carbazole



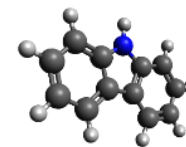
Methane



Toluene



Total Sulfide

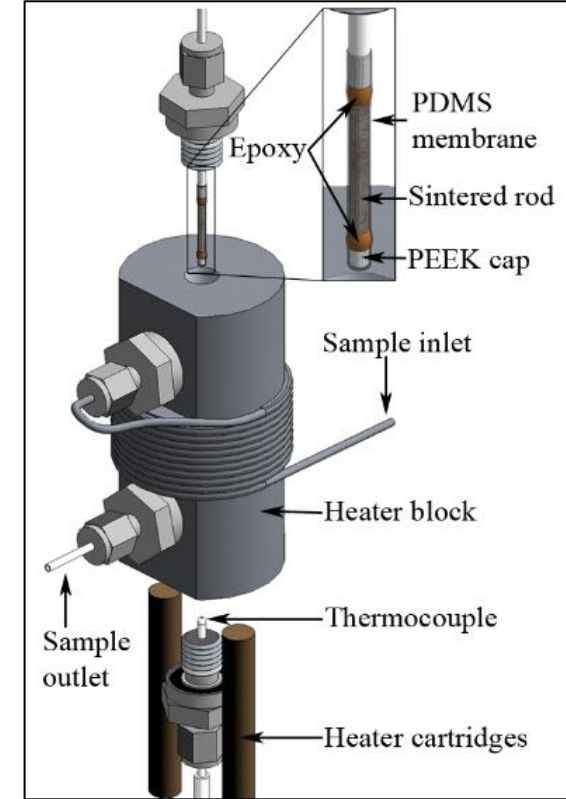
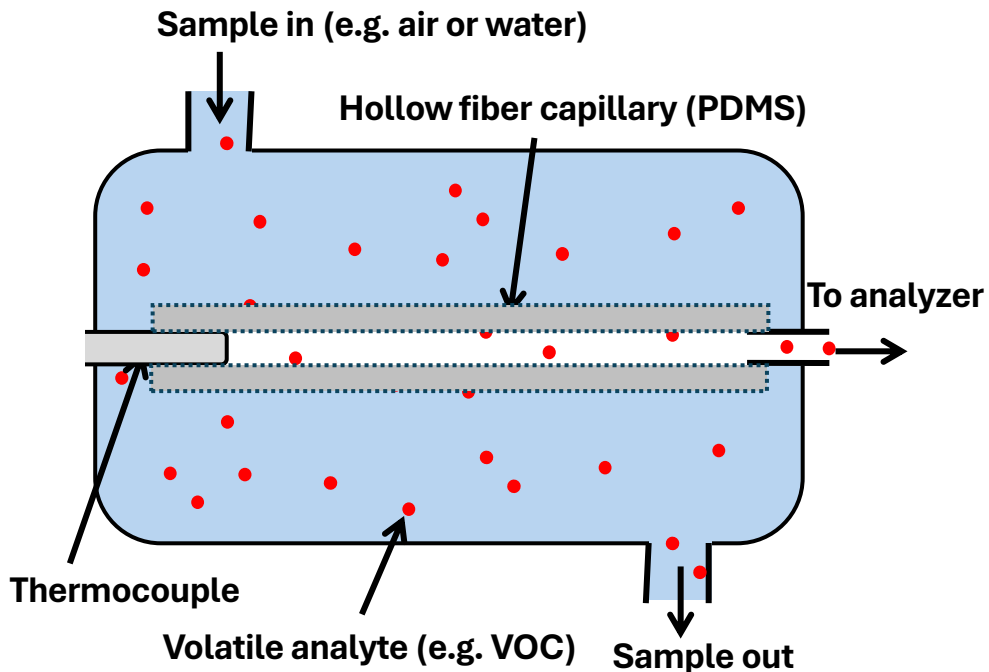


Carbazole



Dissolved Gas Membrane Inlet

Dissolved gases from seawater passed along a membrane's surface diffuse through a polymeric membrane. Vacuum on the membrane's inside then enables rapid gas transport to the analyzer.

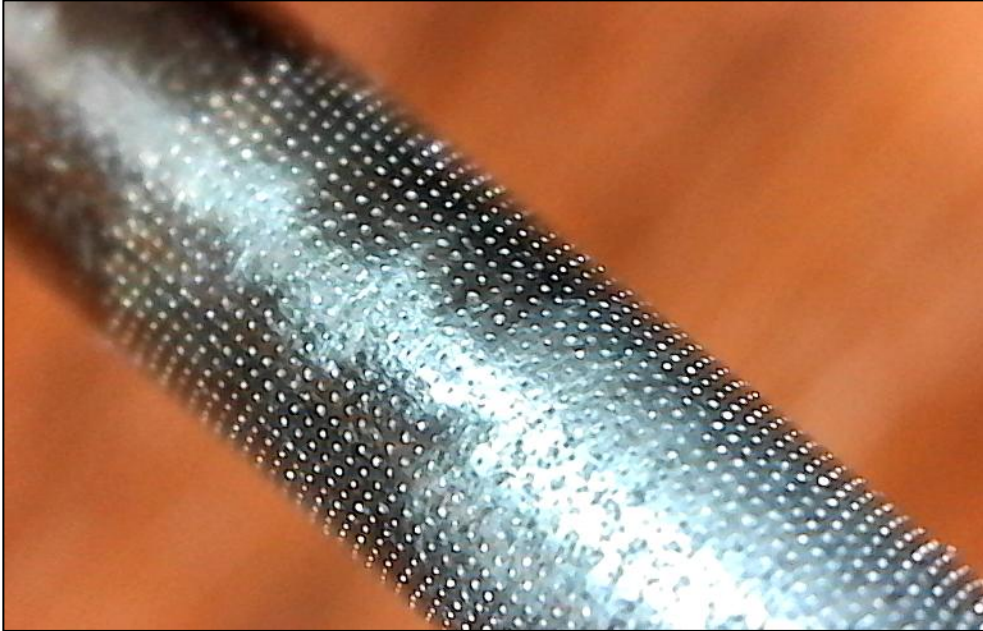


No sample preparation

Incoming sample may be heated for improved analyte response time and provide consistent calibration conditions



Membrane Support



Membrane assemblies

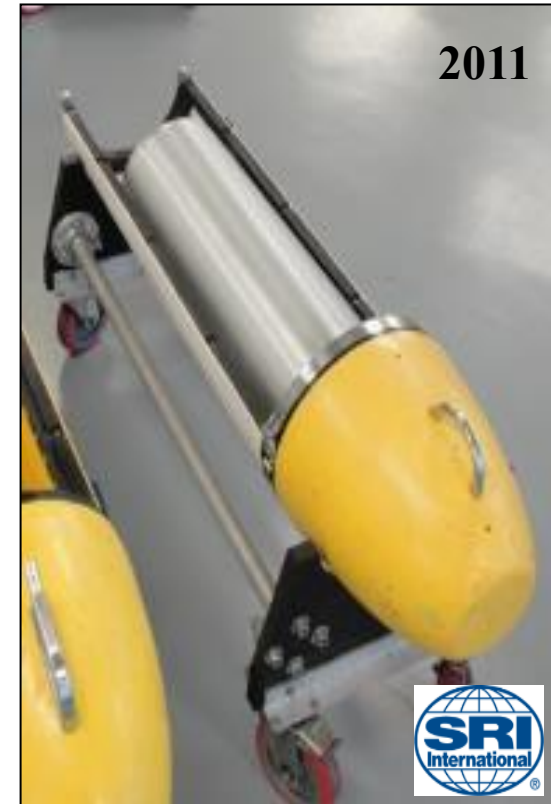
- 7,000 or 10,000 psi max test pressure
- 50 cycles and 1 week sustained max P
- Qualified for 3000 m depth (4500 psi)
- Isolated from vacuum by high P valve
- Various membranes used

We've used many different types of membrane support.
We are currently using this laser-bored steel tubing provided
by Dr. Torben Gentz.



Portable Mass Spectrometry - Past

Predecessor systems at USF and SRI.



Portable Mass Spectrometry - Present

BCA has developed multiple UMS system for the oil and gas industry.

Rated to 3000 m and typically integrated into are larger deployment system for sampling seep-affected seawater

Each system is a custom builds. Systems available for lease.

BCA has been partnered with Dr Torben Gentz and Malte Höhn in Bremerhaven to produce another system.

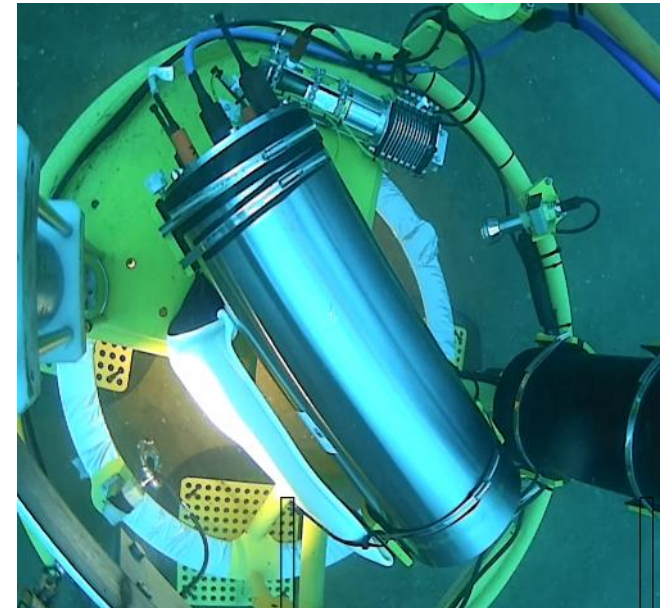


No Cryotrap, external
flow control

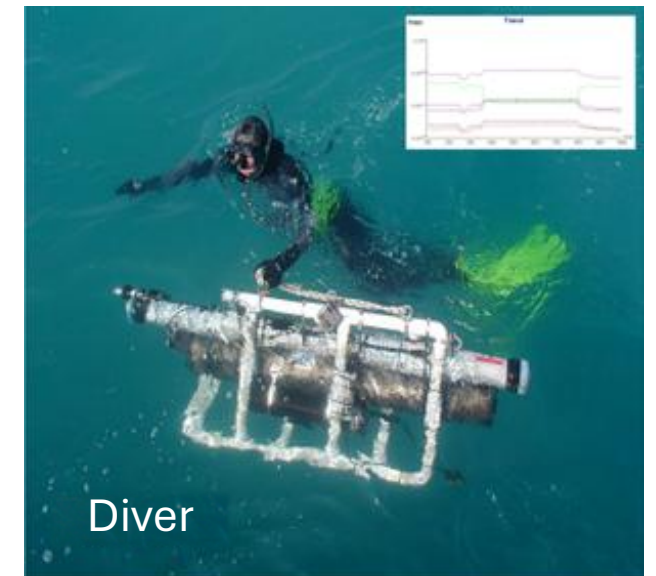
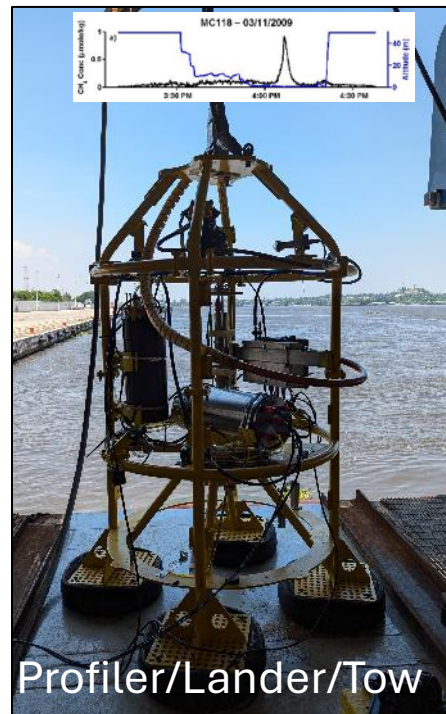
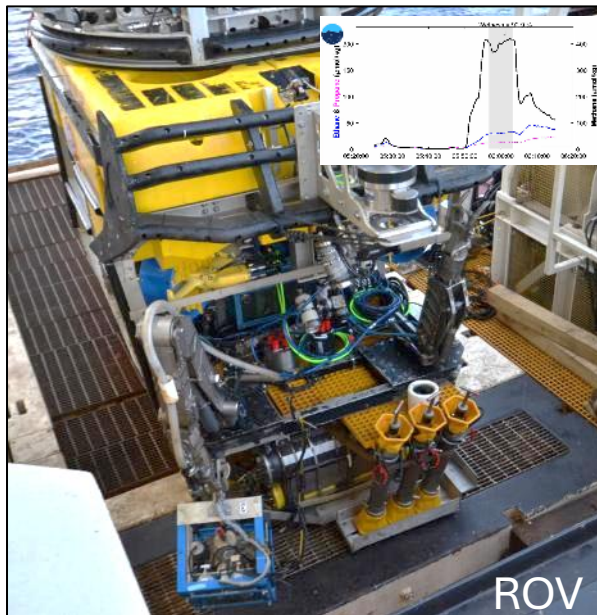
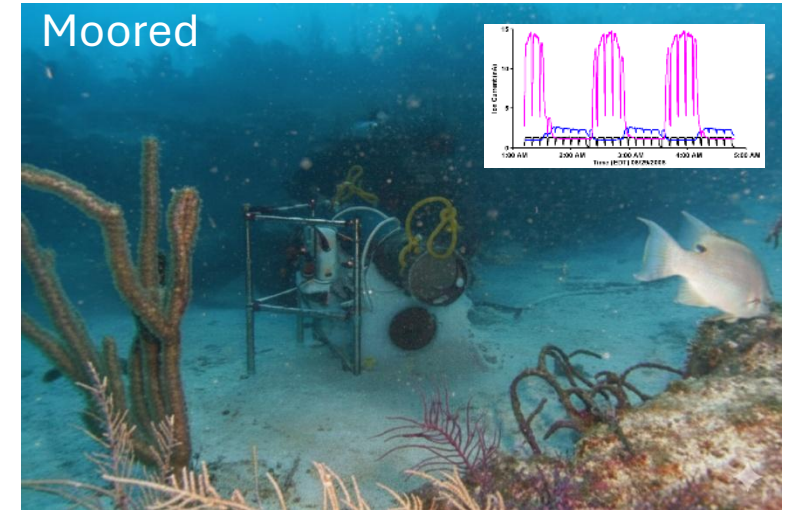
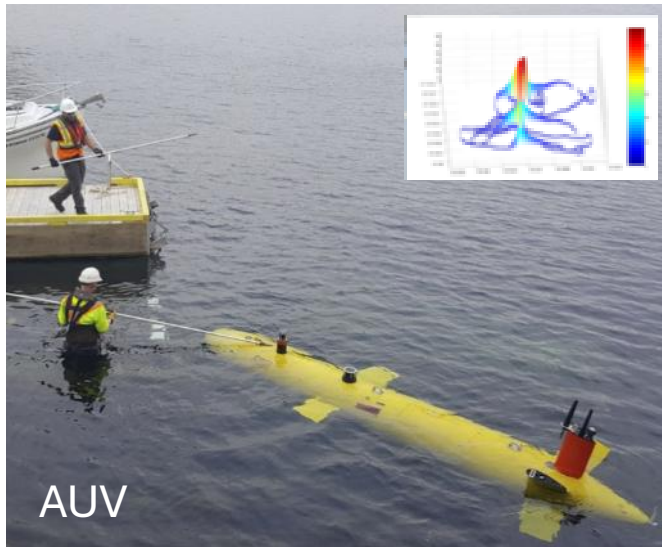


Cryotrap, internal
flow control

Cryotrap, external
flow control

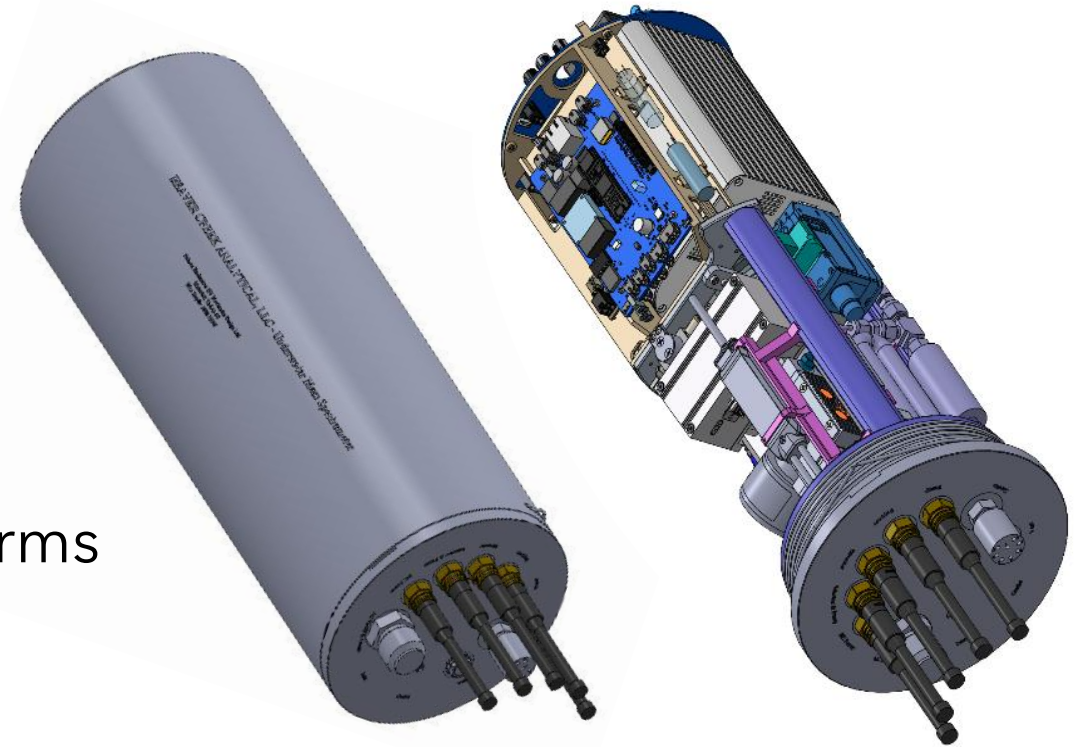


Platform Independent

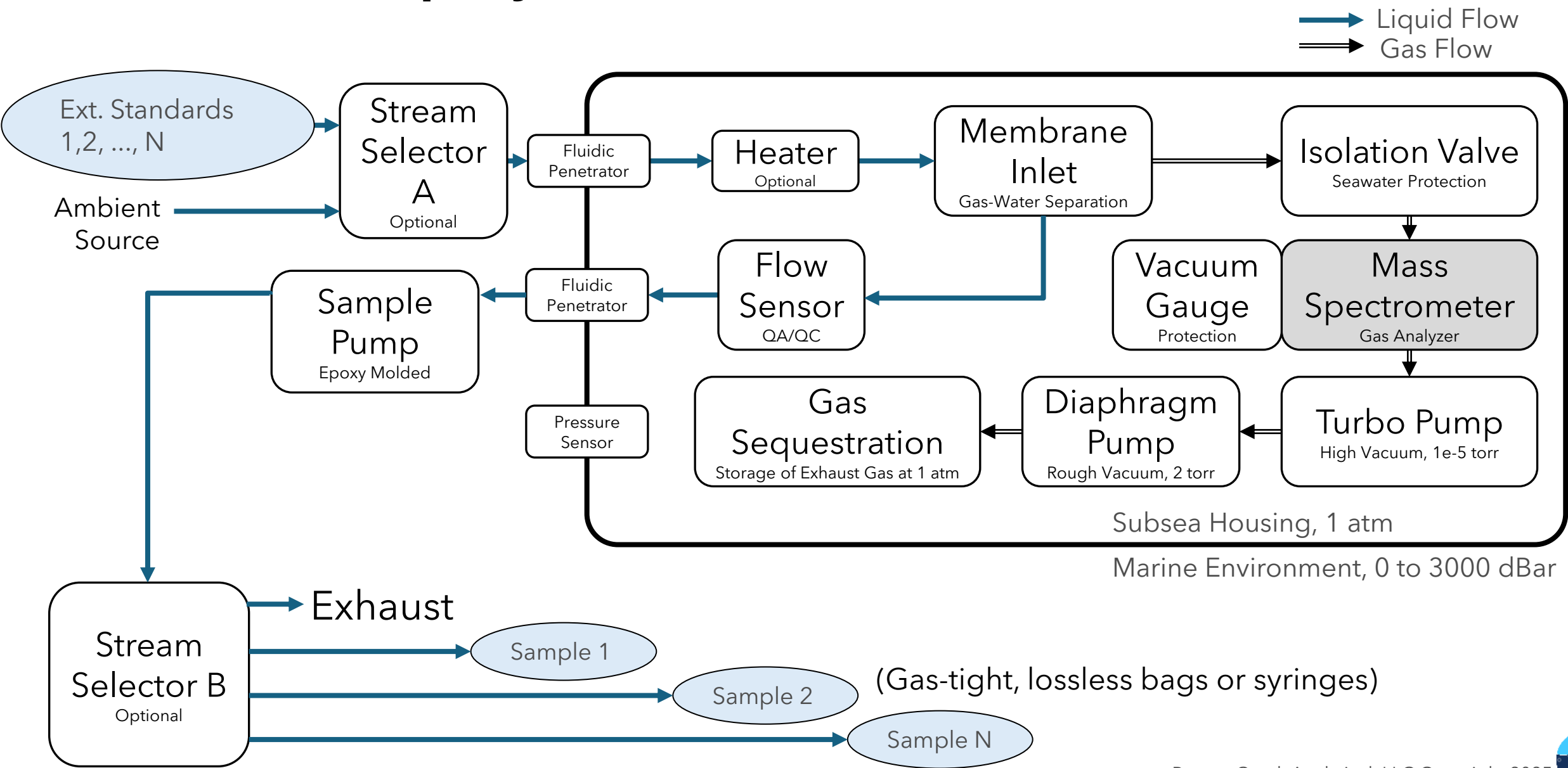


Operational Specifications

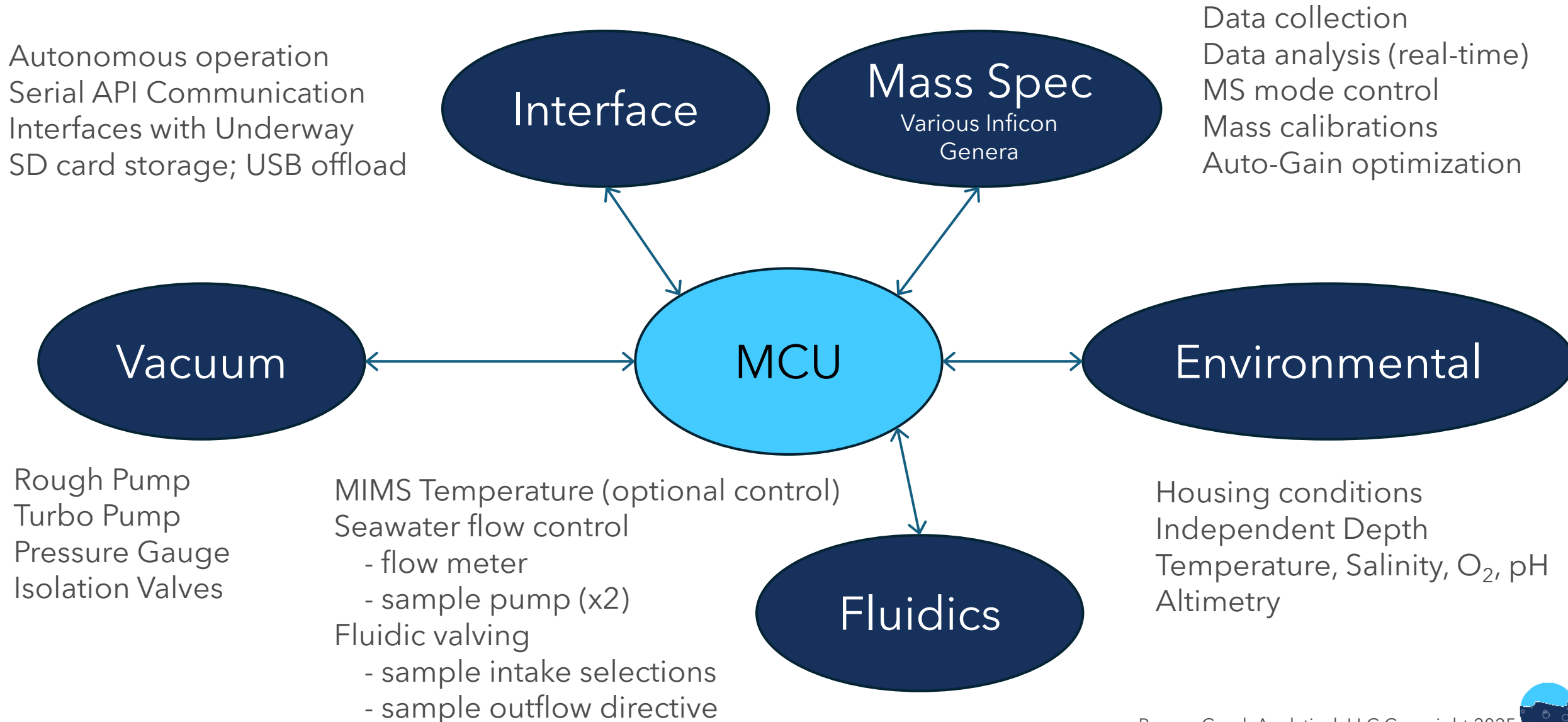
- **Depth Rating:** 3000 m
- **Power:** 65 to 80W
 - Primary Input: 110 - 250 VAC
 - Secondary Input: 18 to 60 V
 - Hot swappable power between platforms
- **Diameter:** 202 mm (8 in)
- **Length:** 526 mm (20.7 in)
- **Weight:** 34 kg in Air, 17 kg in seawater (MSW)
- **Communication:** Serial (57600 baud, 8N1), internal data storage
- **Networking:** 12 W supply and RS-485 from the UMS enables networking with additional sensors



Deployable Schematic Overview



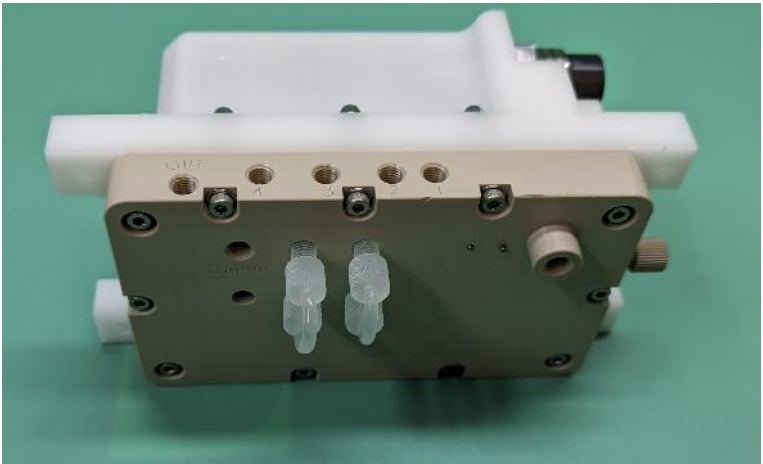
BCA UMS Microcontroller



Subsea Valves for In Situ Analytics

Multi position valve apparatus can be plumbed according to research needs:

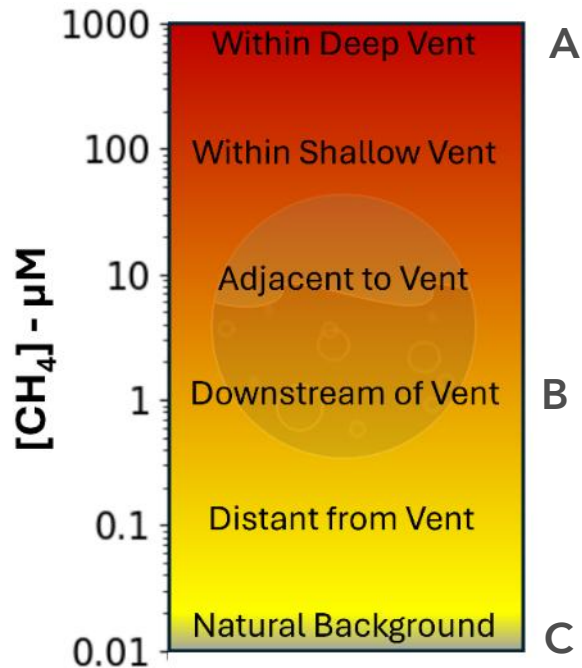
Valve Task	Consequence
Inlet-to-sensor timing calibrations	Timing offsets for mapping corrections on fast moving vehicles
Sensor-to-sampler timing calibrations	Sensor-triggered sample collection for accredited laboratory analysis
UMS sensitivity calibrations	Baseline, sensitivity and detection limit
Recirculation	Steady-state measurements



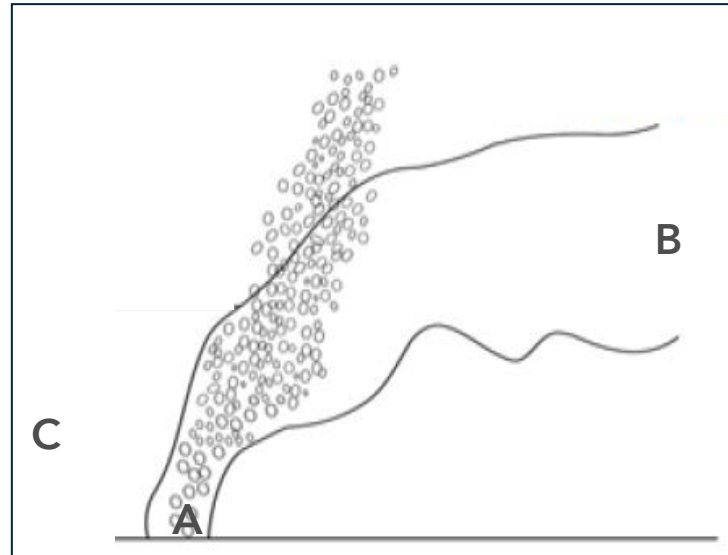
BCA 4-position subsea valve with internal sample pump

Analytical Specification - Detection Limits

Excellent detection limits for dissolved gases (by 3σ at background)



Typical Concentrations



Methane - <20 nM (0.3 ppb)

Ethane - 30 nM (0.9 ppb)

Propane - 30 nM (1.3 ppb)

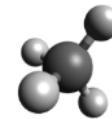
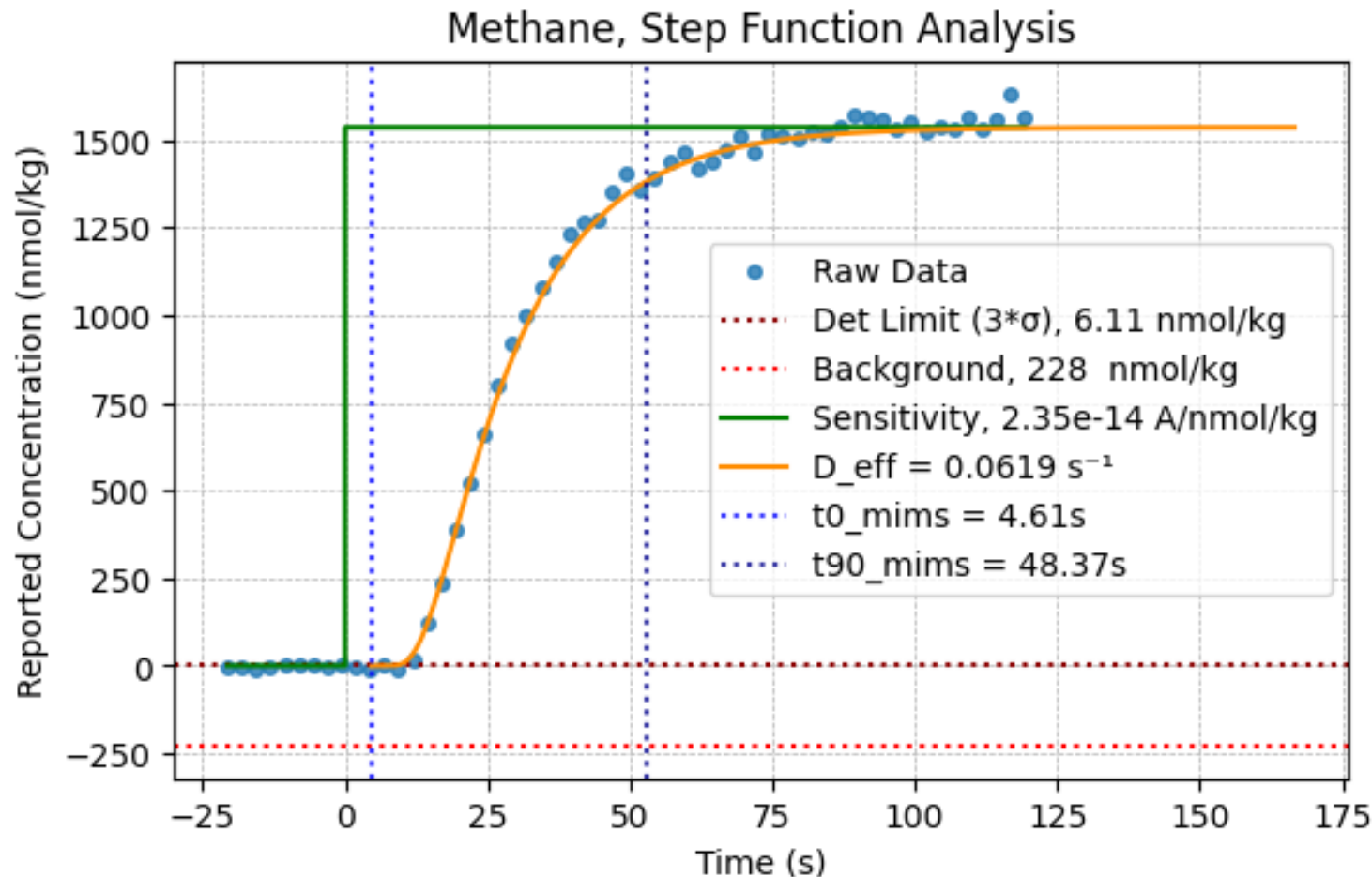
Butane - 30 nM (1.7 ppb)

Detection Limits



Analytical Specification - Response Time

Excellent response times, $T_{90} = 20 - 60$ s for most analytes



Methane



UMS Mode Selection

- User can rapidly switch between UMS modes:
 1. Survey Mode – Rapid, quantitative detection for C₁-C₄
 1. Optimized for methane detection limit (about 2 nmol/kg)
 2. Extremely low interferences for methane
 3. Very rapid scan rate and high sample flow rate
 2. Extended Mode – Semi-quantitative detection
 1. Optimized for VOC detection limits
 2. Full scan picks up unexpected analytes
 3. Heated membrane

Semi-Quantitative Analytes

Cyclopentane

Cyclohexane

Pentane

Benzene

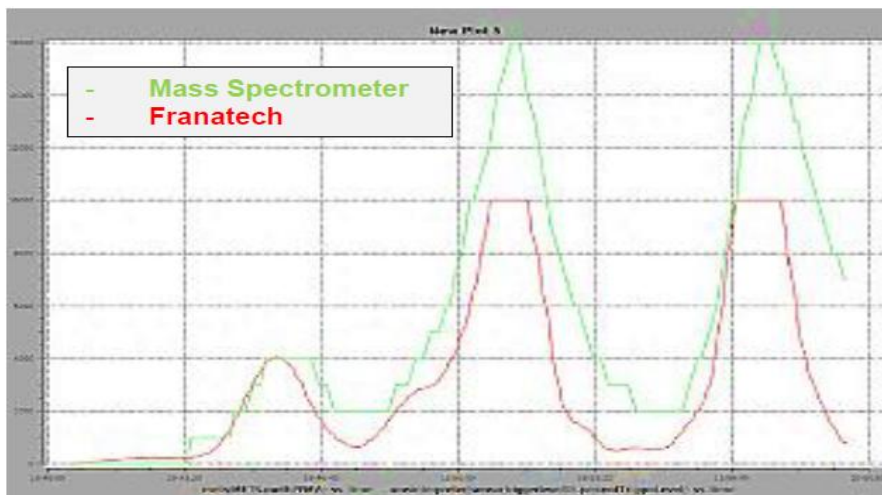
Toluene

C5+ Alkanes

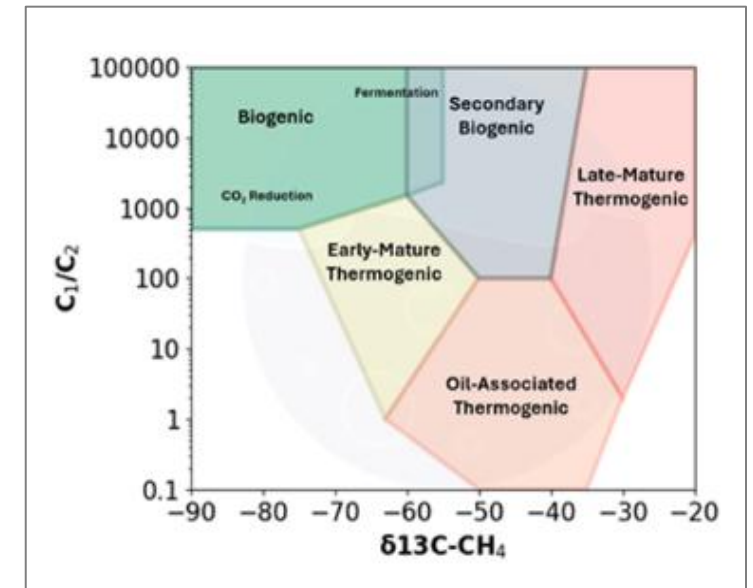


Analytical Specification - Linear Dynamic Range

- The UMS has a linear dynamic range of $\times 10^6$
 - Map distant plume with low DL
 - Obtain ratiometrics from within plume source
- With simultaneous detection of multiple analytes, collection ratiometric data with vastly different concentrations is enabled
- This is where the mass spectrometer shines



Plume data from Hechanova, ITAC 2016



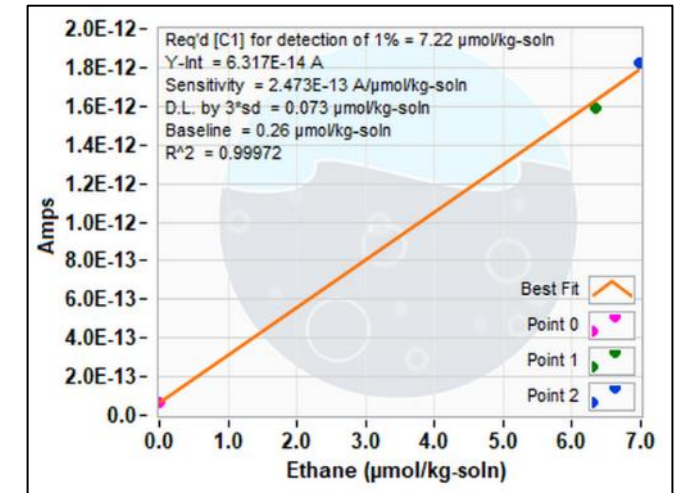
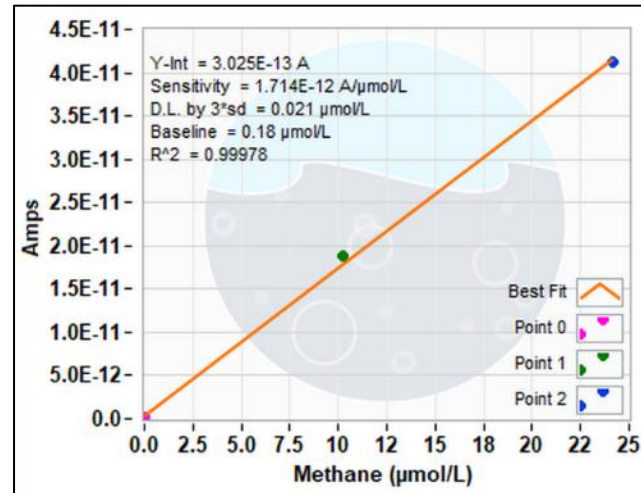
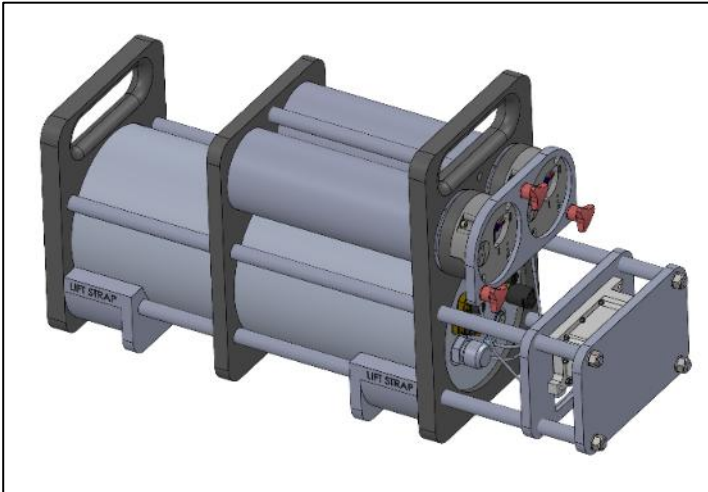
Bernard diagrams covers 6 orders of magnitude for C1/C2



Calibration Approaches

- **Qualitative** → low effort
- **Semi-quantitative** → ship-board calibrations, corrections
- **Quantitative** → in situ calibrations, subsea fluidics

Hardware and techniques to accomplish each of these objective have been deployed



Cryotrap

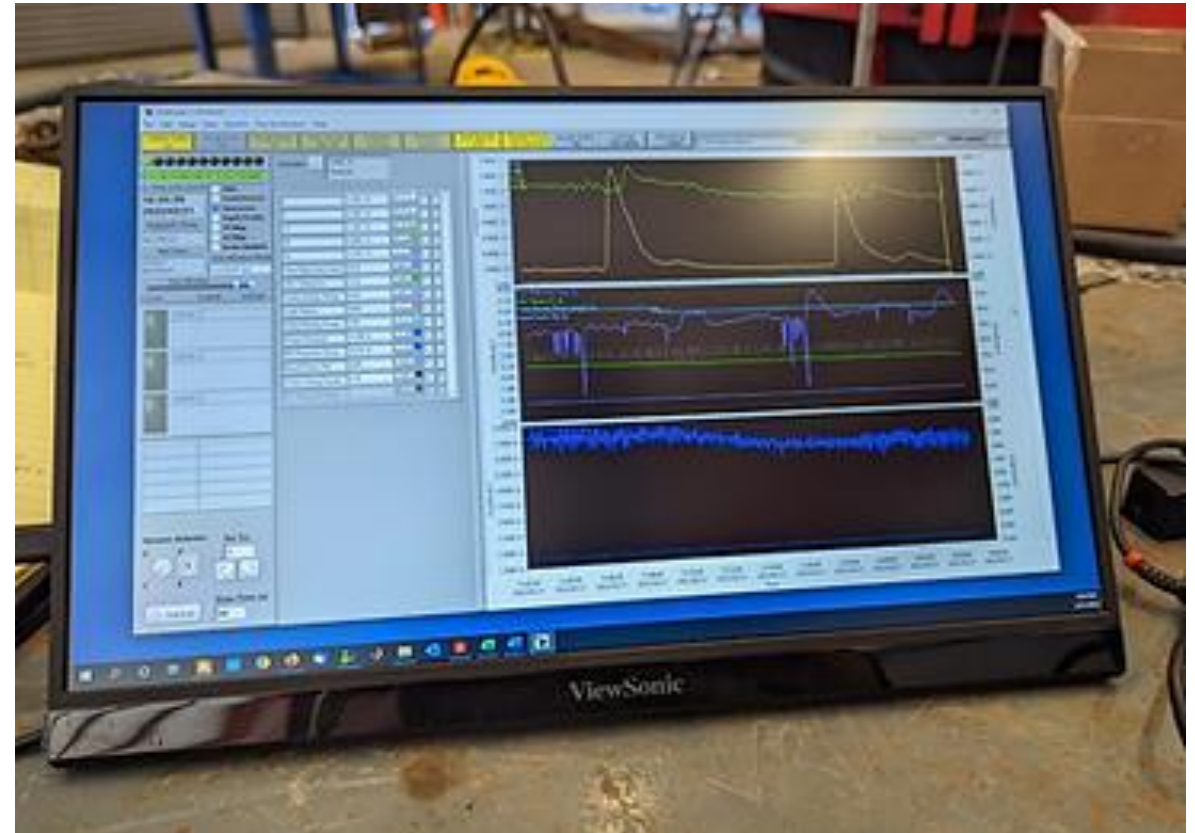


- Stirling pump, at a few watts, dramatically reduces instrumental noise by trapping water vapor between the membrane and analyzer
- Requires periodic regeneration
- Enables further improvements in detection limits
- Dramatically extends filament lifetime



Operational Software - Underway

- Control interface for UMS and additional sensors
- Data logging and display for 10 additional sensor
- 3-Graph, 6 y-axis timeseries for plotting data and metadata
- Report generation



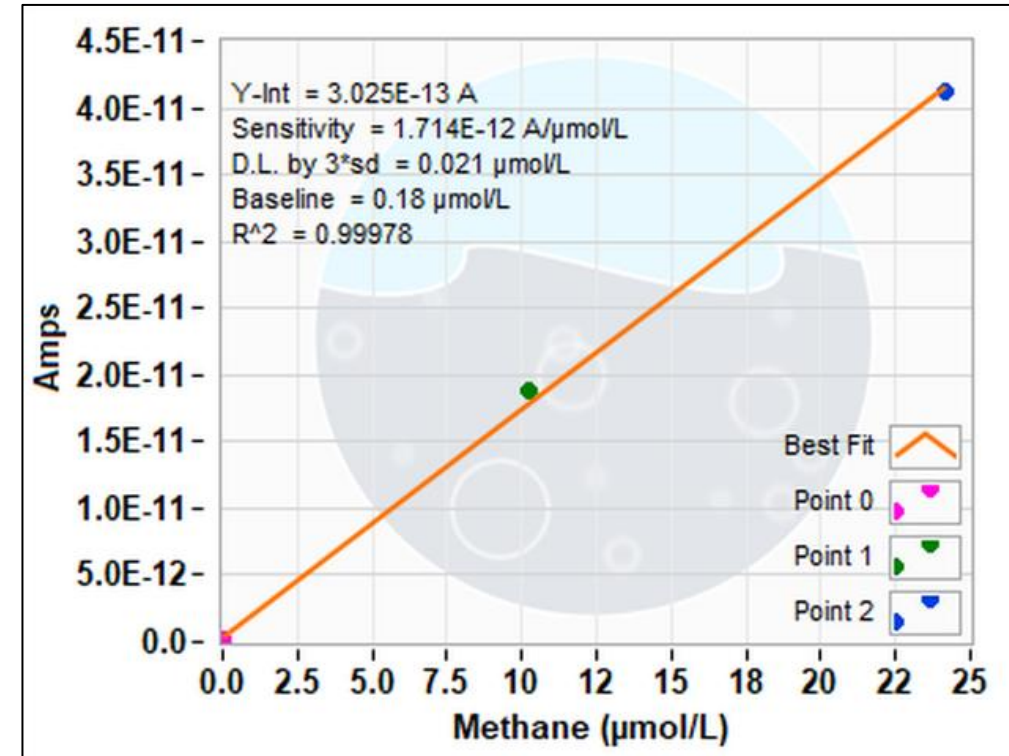
Multi-sensor timeseries



Operational Software - Underway

- Calculation of custom mathematics, seawater properties and coordinates system transformations
- In situ, dissolved gas calibration features
- Real-time UDP data transmission

Raw MS output

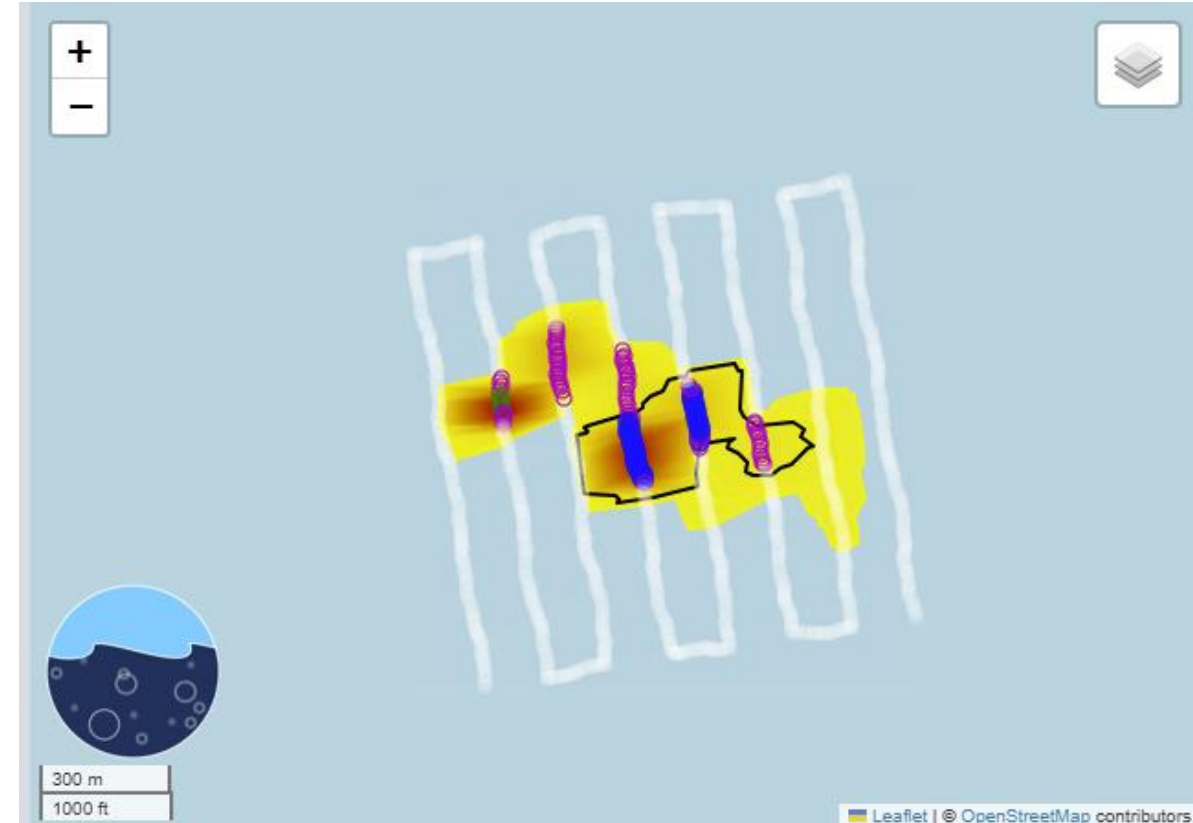


Multi-point, on-site, in situ calibration



Operational Software - Underway

- Real-time 2D and 3D mapping
- Export of timeseries and maps as self-contained, interactive html file

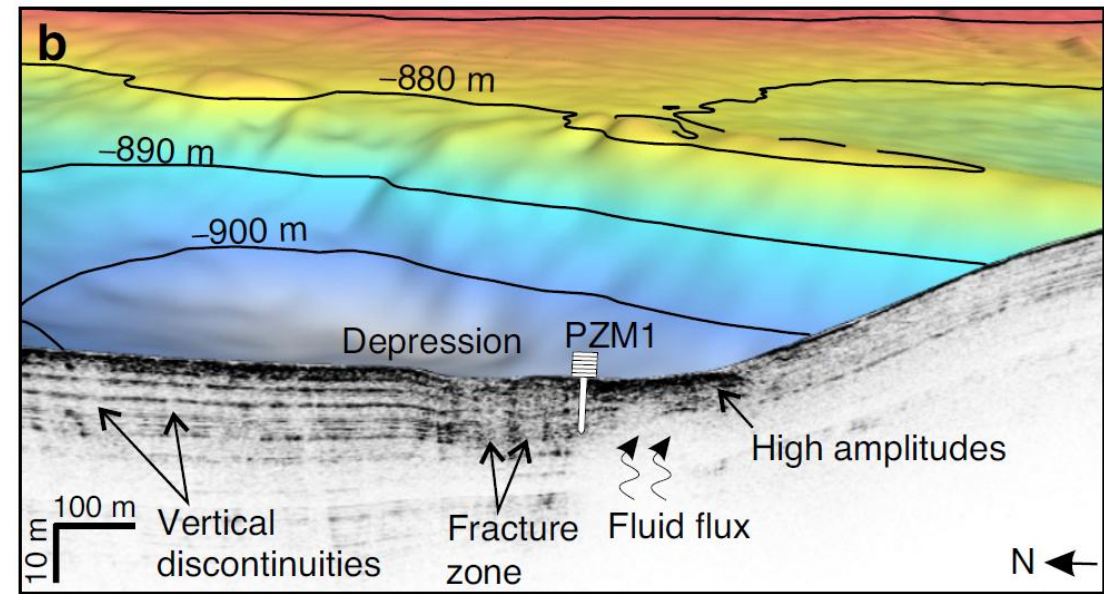


Seeps Description



Cold Seeps - Occurrence

- There are numerous mechanisms through which cold seeps containing oil and gas are known to occur
- One mechanism particularly relevant to many offshore formations is hydraulic fracturing
- Typical features:
 - Complex geologies
 - Salt diapers
 - Brines
 - Mud volcanoes
 - Hydraulic pressure and fracturing
 - Tectonic fracturing
 - Benthic communities
 - Buoyancy forces
 - Hydrate formation



Sultan et al, 2020



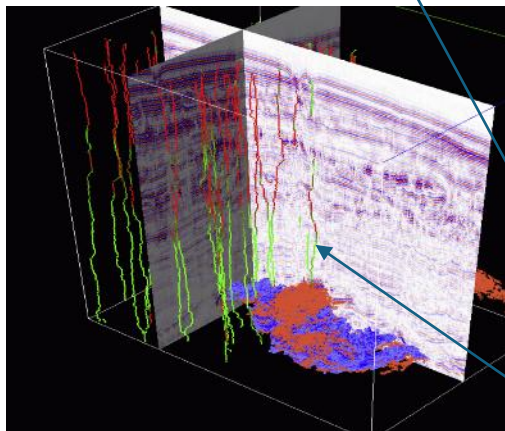
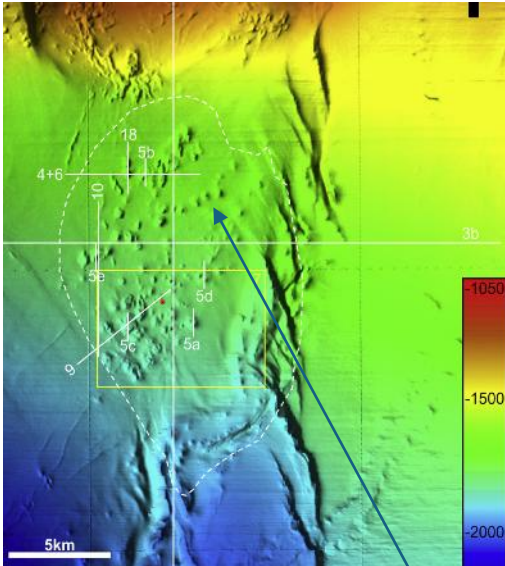
Cold Seeps - Hydraulics

1. Oil and gas reservoirs establish an overpressure:

1. Rapid or lateral compaction
2. Hydrocarbon Generation
3. Thermal expansion

2. Reservoir overpressure is relieved by hydraulic fracturing which leads to Pipe/Channel formation and blow-outs

3. Continued, steady, **seepage** through blowout channels relieves the continued development of reservoir overpressure



Graue et al,
2011

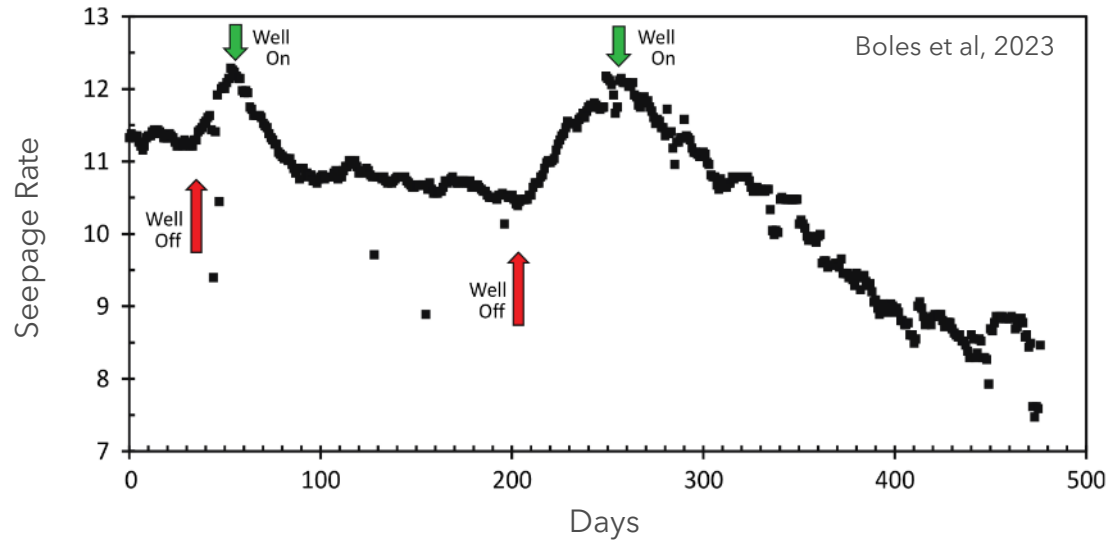
Blow-out pockmarks located by multibeam and pipes located by seismic



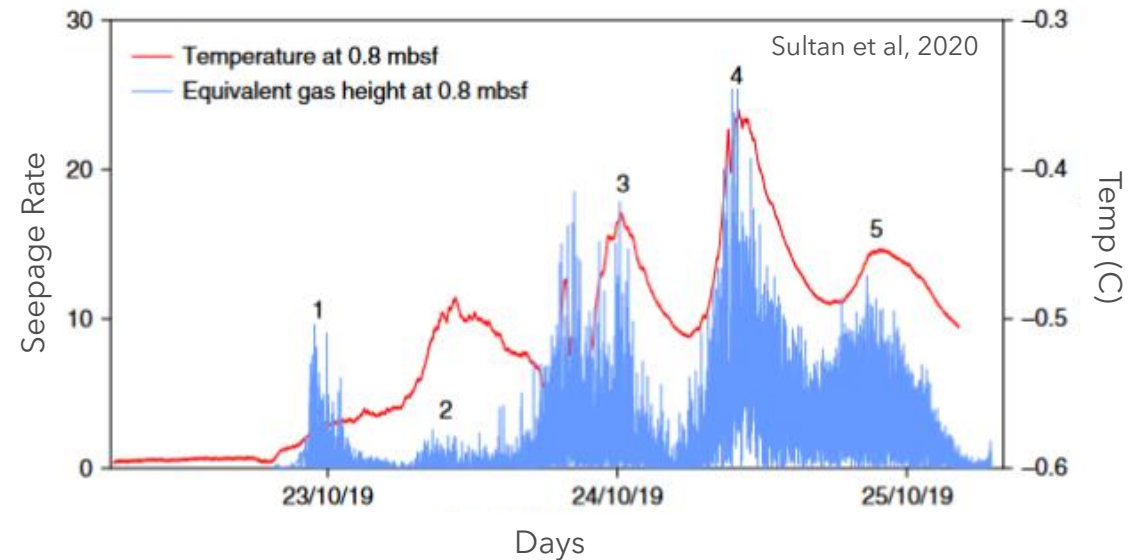
Cold Seeps - Occurrence

- Hydrocarbon seepage rate can be controlled by hydraulic pressure. Seepage can be reduced through
 - Well-head draw-down (e.g. extraction)
 - Top-side pressure changes (e.g. tides)

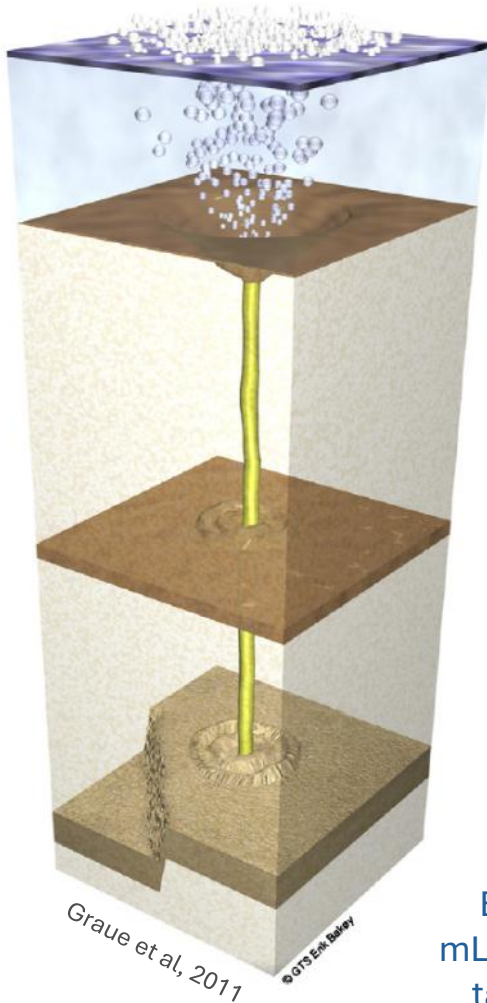
Hydraulic link between extraction and seeps



Hydraulic link between tides and seeps



Cold Seeps - Occurrence



Seep fluid, driven rapidly to the surface through established pipes, provides a direct link to reservoir conditions.

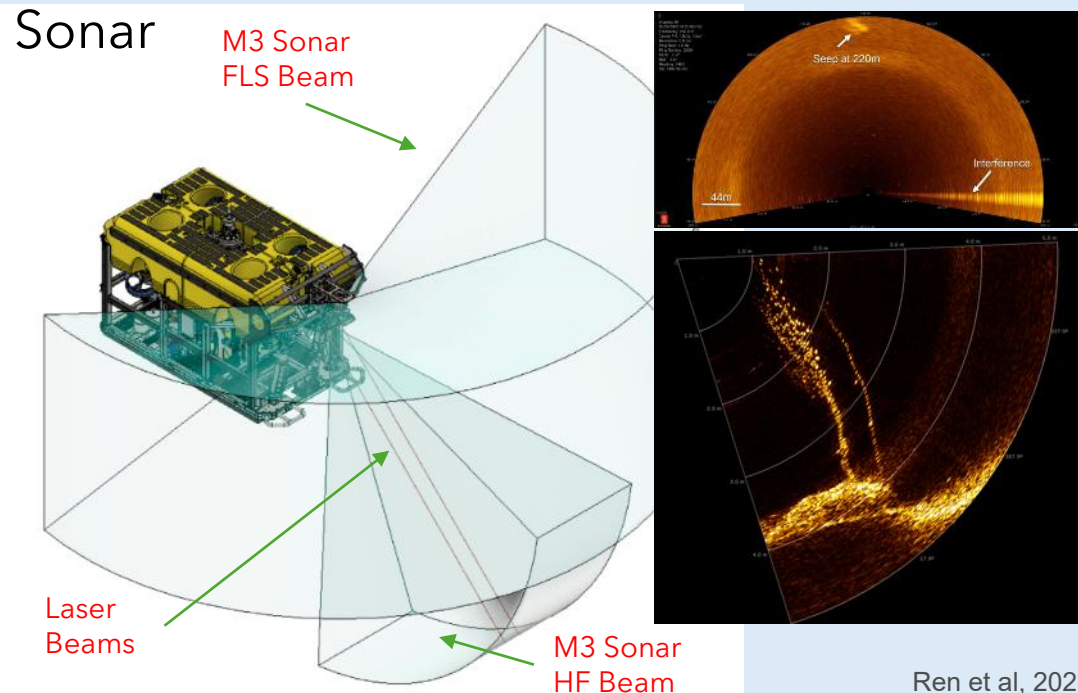
Under the right conditions, direct analysis of **active** seep fluids enables **characterizations of reservoir fluids**.

By our math, a fluid flowing at 1 mL/min through a 10mm^2 vein could take about **7 days** to travel 1 km (e.g. reservoir to seafloor)



Cold Seeps - Tracking

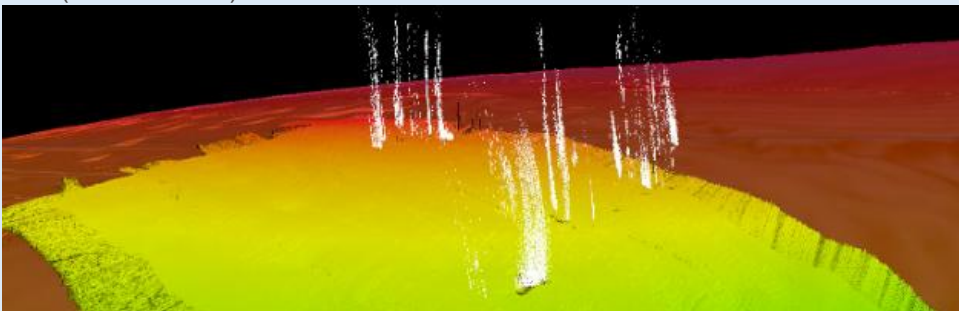
Sonar



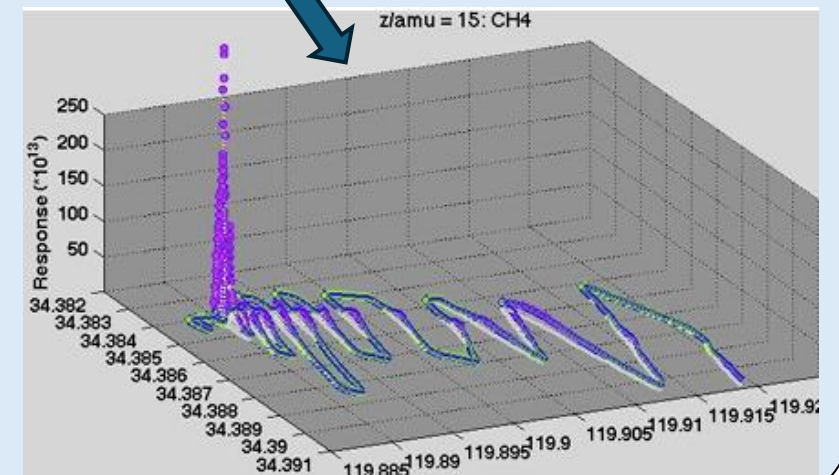
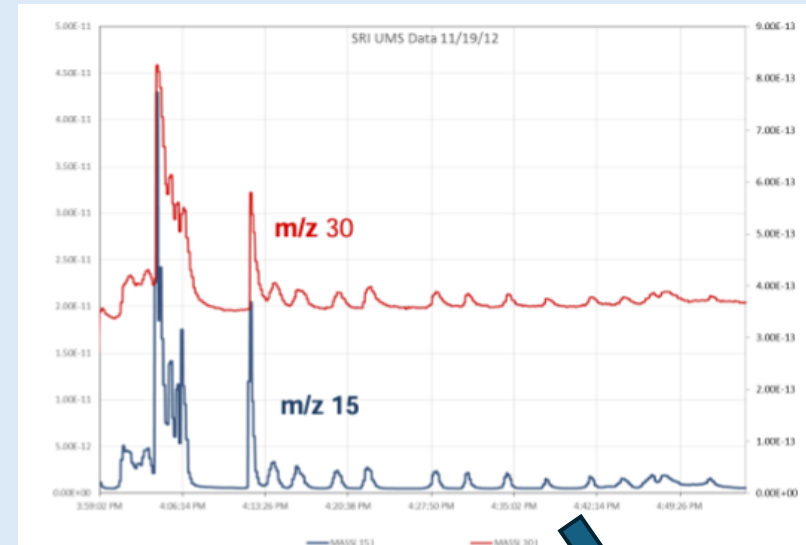
Ren et al, 2023

NSF (Adam Skarke)

Multibeam

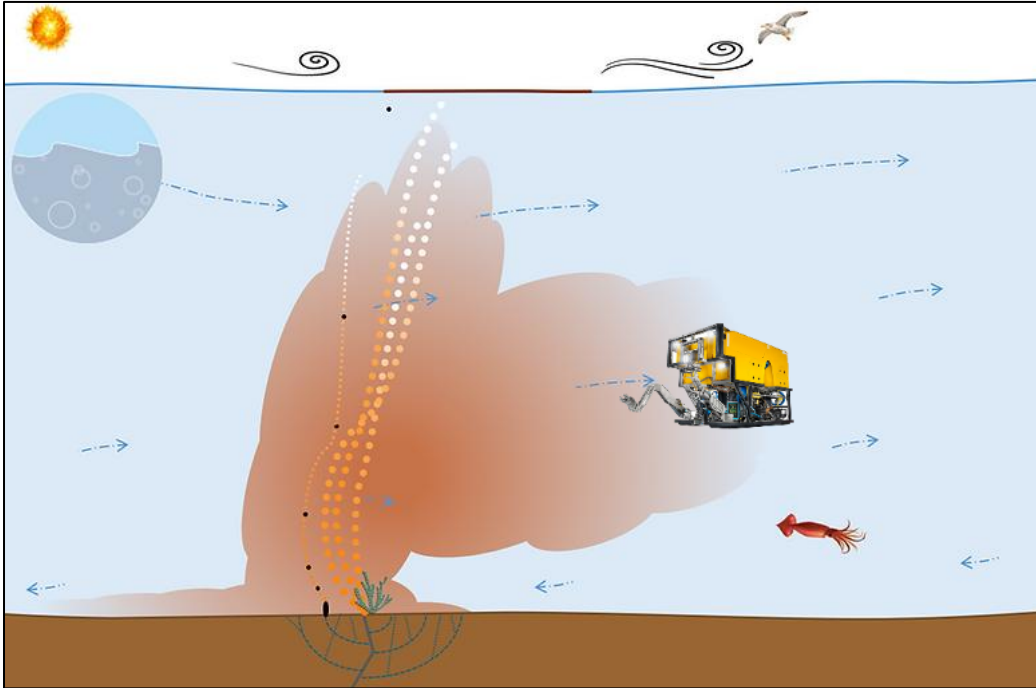


Chemical



Short et al, 2015

Cold Seeps - Characterization

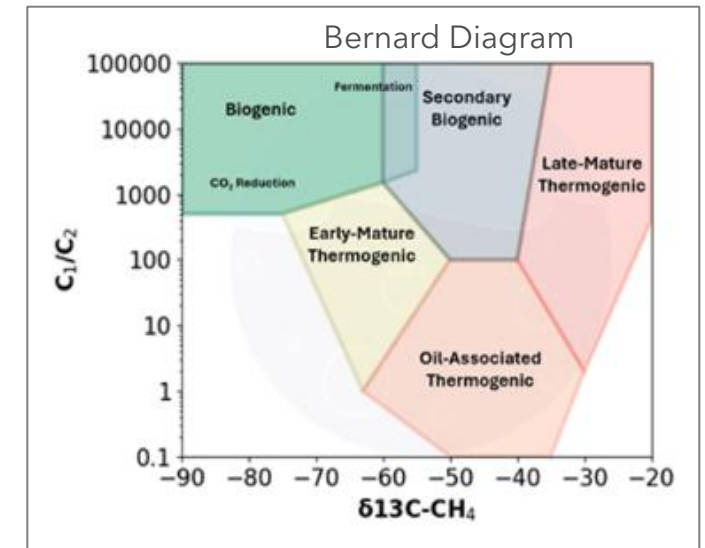


Survey Area

- Detection of CH_4 and possibly C_2H_6
- Establish likelihood of thermogenic with C_1/C_2 ratio.
 - $\text{C}_1/\text{C}_2 < 20$ - likely thermogenic
 - $\text{C}_1/\text{C}_2 > 100$ - likely biogenic
- UMS-triggered sample collection for accredited ex situ analysis (full composition and isotopes)

Extended Analysis

- UMS detection of certain light hydrocarbons indicate the seep fluid is 'live' or 'unwashed', suggesting a direct-to-reservoir pathway
- Collect visual data, bio-data, seep fluids, and cores
- The precision of an ROV can avoid degraded oil in the near-seep vicinity and target the active seep fluid



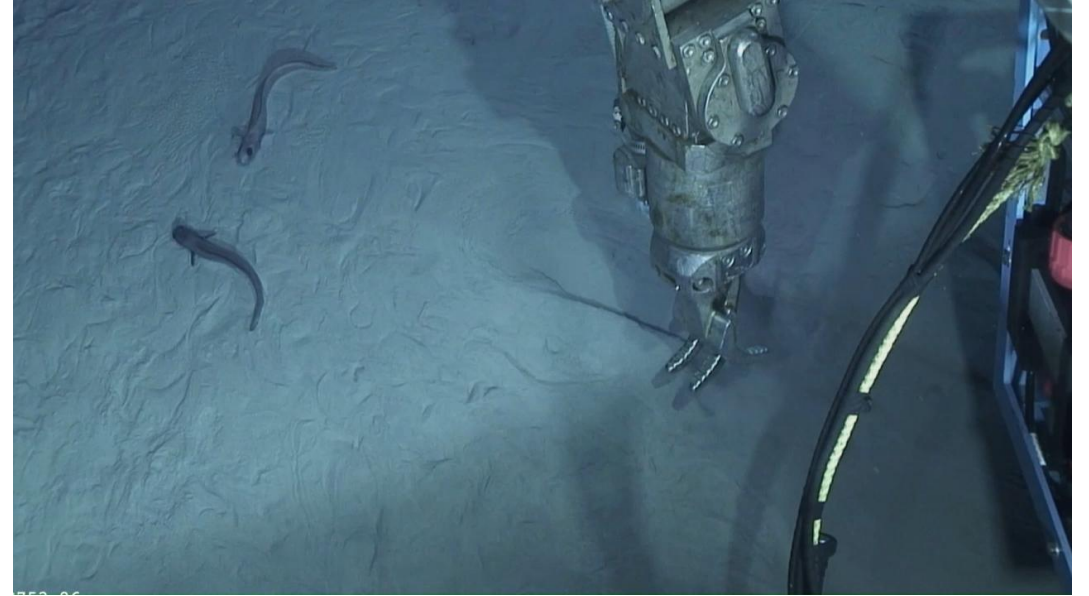
Cold Seeps - Variety



Oily Seeps



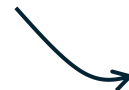
Mud Volcanoes



Gassy Seeps



Sublittoral Seeps



Cold Seeps - Variety

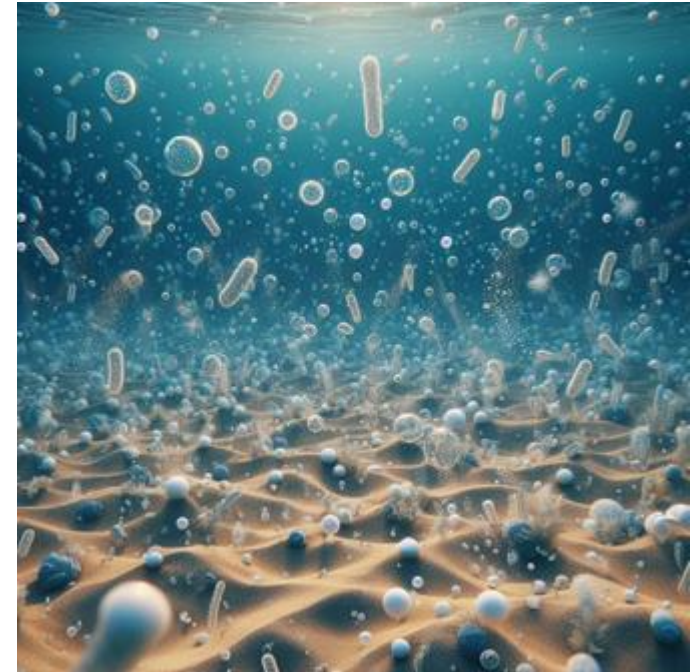
Subsea brine pools



Methane in the Water Column is Dynamic

- Methane sources in the ocean are **limited** and **localized**
 - Rivers, seeps (biogenic, thermogenic)
- Major sinks in the ocean are **strong** and **vast**
 - methanotrophic bacteria
 - air-sea interaction

As a result, methane concentrations are extremely transient and are reduced to less than 10 nM (0.16ppb) very rapidly



Quantifying Dynamic Hydrocarbons Ratiometrically

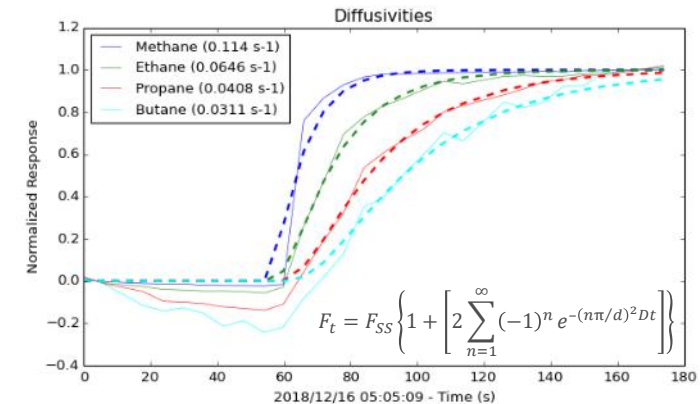
- **Problem**

- The UMS produces methane, ethane, propane, and butane data each with different response functions
- Directly comparing one to another ratiometrically is challenging and error prone

- **Customer Solution:** Recirculate the sample at the UMS inlet and perform analysis for an extended period.

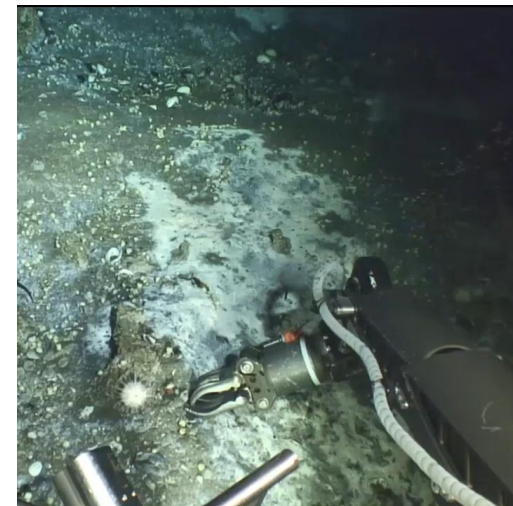
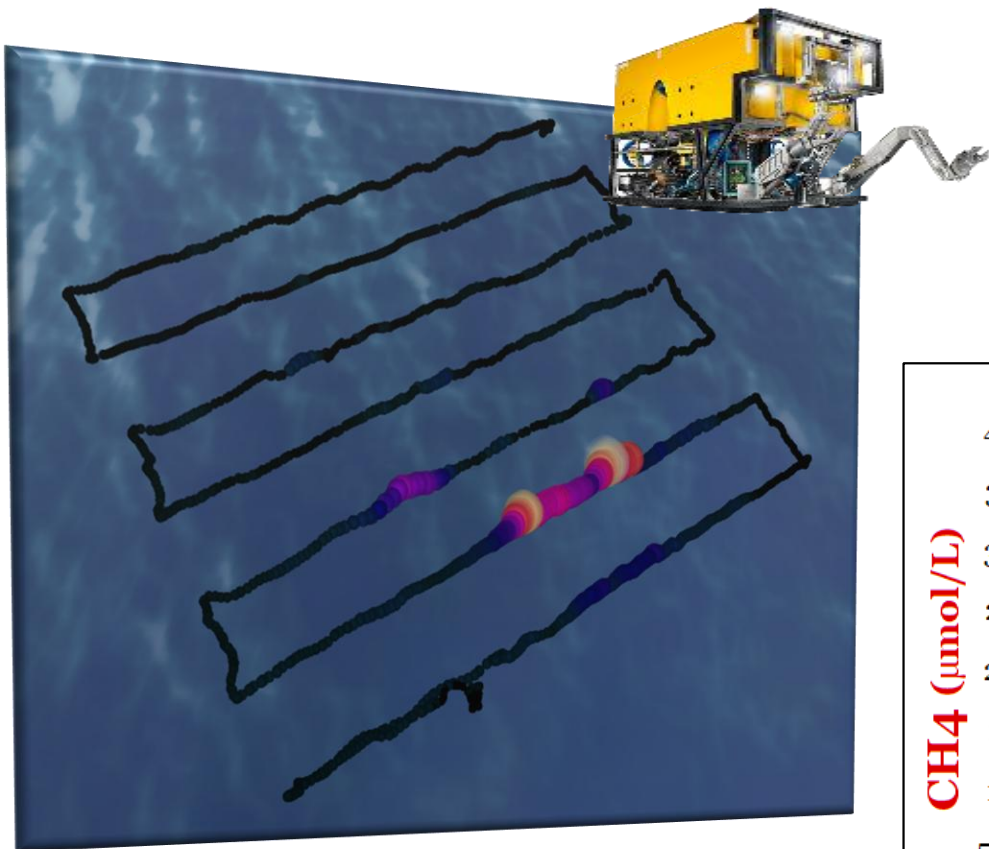
Ratiometric data is wild during survey-mode analysis

Ratio data steady when steady state analysis is achieved

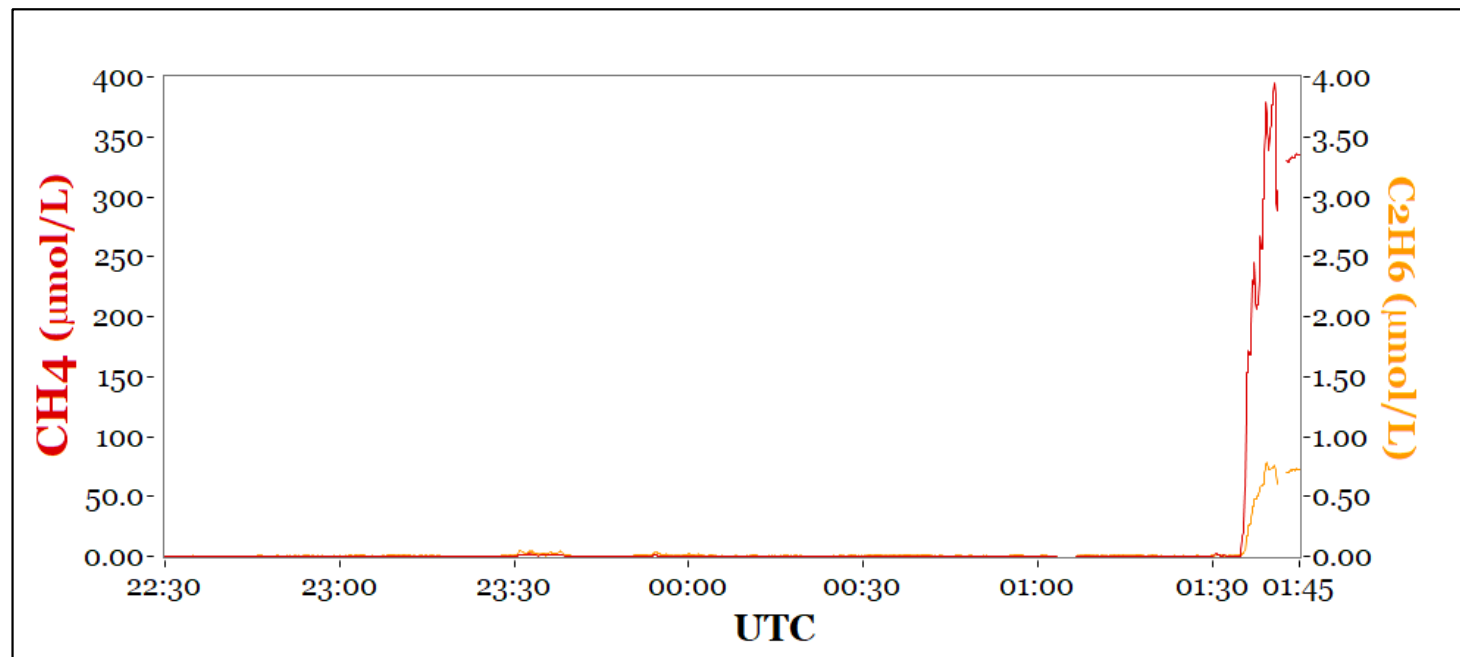


Cold Seeps - Approach

Methane and ethane exploration over known vents (1100 m)



At Seep: C1/C2 accurately determined,
and VOCs detected



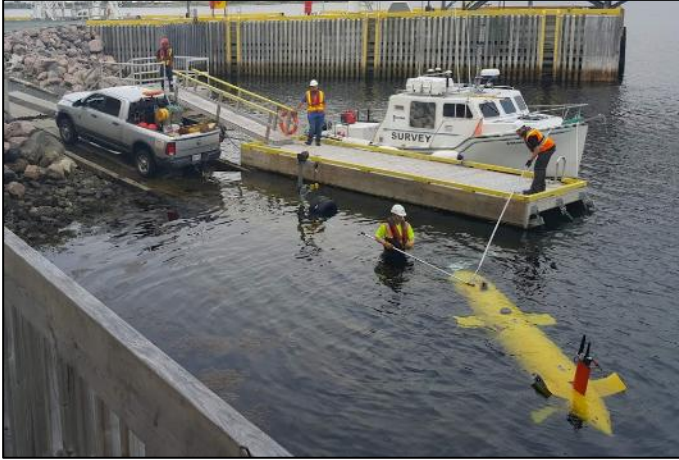
Survey: Seep discovered (sonar) and
C1/C2 ratio roughly establish



Results

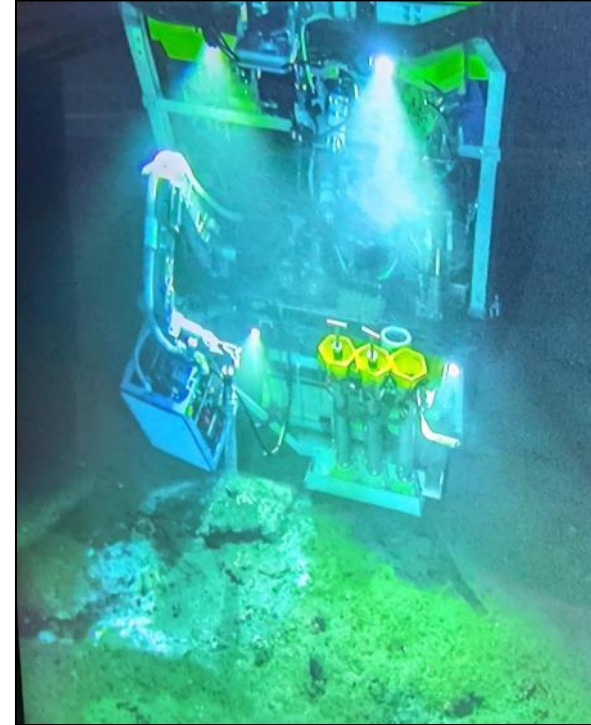


Commercial UMS Usage



Seep plume mapping by AUV

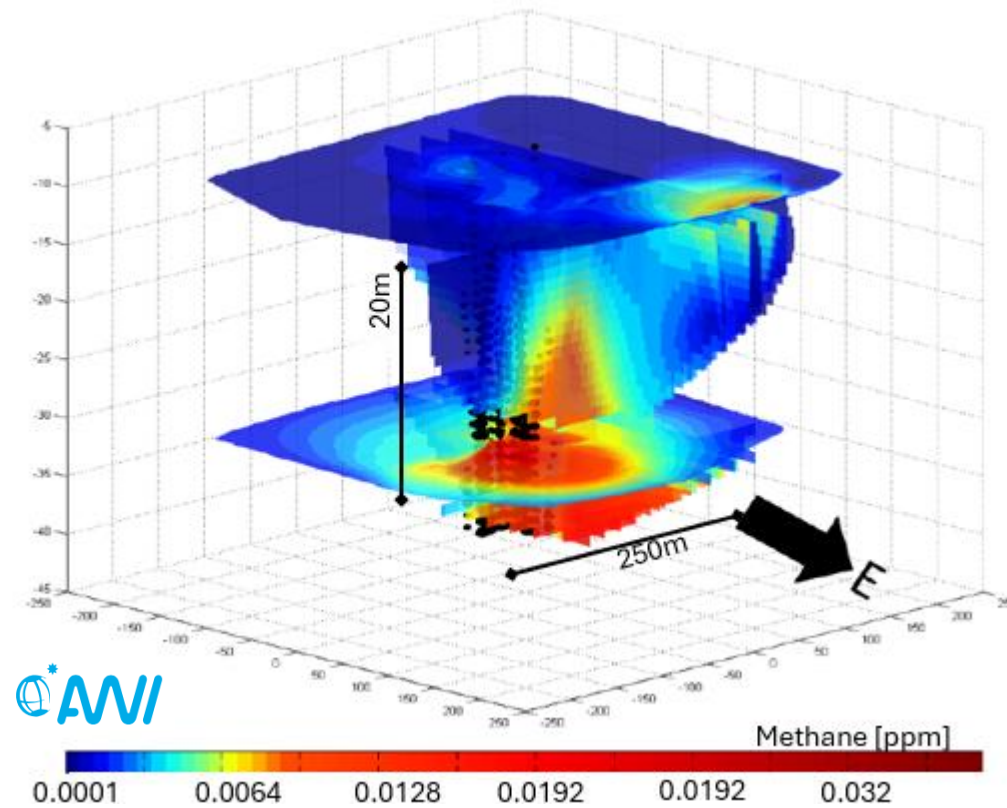
>150 seeps characterized by ROV over multiple cruises



Seep characterization and broad area survey by lander (over 500 sites surveyed via 24 hr operations)



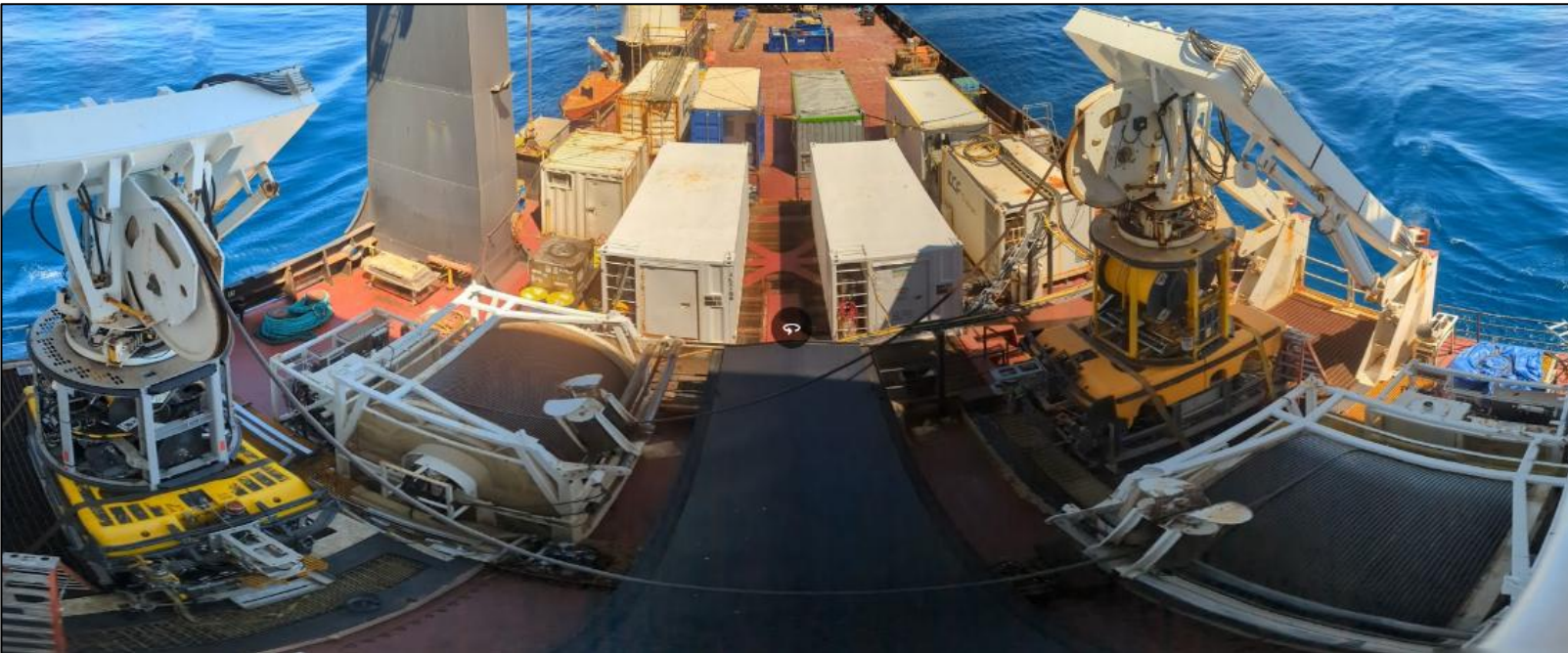
Plume Mapping by AUV



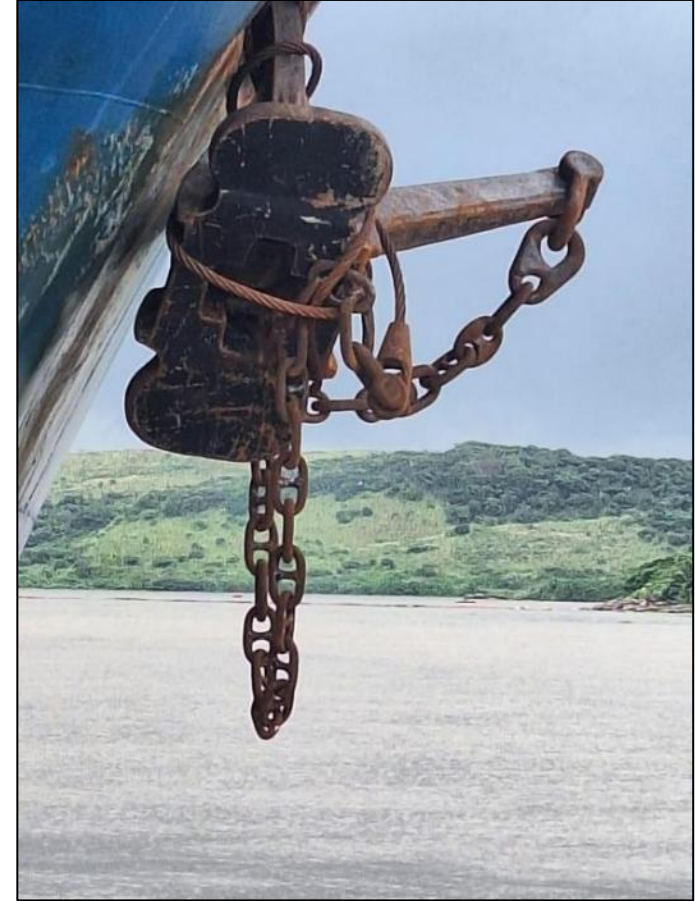
Seep Characterizations by ROV



- On board control center
- 24 hr operations
- 2 to 4 weeks at sea
- Multiple operator training required



I open the door to questions...



...and hope to untangle them.



www.bcanalytical.com

